

STATISTICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: Percentiles

Grading guide. Total 22 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Line them up

1. Nine preemies weigh (in grams): 1,100, 760, 940, 1,600, 880, 990, 820, 1,250, 900. (a) Write them in order, smallest to biggest. (b) What is the median? (c) The mean is about 1,027. Why is the mean higher than the median here? (5 pts)

Answer: (a) 760, 820, 880, 900, 940, 990, 1,100, 1,250, 1,600 (2 pts; 1 pt if one swap). (b) Median = 940, the fifth in line (1 pt). (c) The 1,600 g baby is an outlier that drags the mean up; the median just counts positions, so it stays in the middle of the line (2 pts).

Part B — Place in line

2. Twenty babies are lined up by weight. Turn each baby's position into a percentile. (Position $\div 20 \times 100$.) (4 pts)

Answer: A: $2 \div 20 = 0.10 \rightarrow$ 10th percentile. B: $10 \div 20 = 0.50 \rightarrow$ 50th (the median). C: $15 \div 20 = 0.75 \rightarrow$ 75th. D: $19 \div 20 = 0.95 \rightarrow$ 95th. 1 pt each.

3. Which of these is TRUE about percentiles? (1 pt)

Answer: A. In any healthy group of babies, one in ten is at or below the 10th percentile. - A percentile is a place in line, so somebody is always 10th from the bottom. Small isn't sick; a change is what matters.

Part C — On the chart

4. Plot these three babies on the chart as dots with their letters. Baby X: born at 28 weeks, 1,150 g. Baby Y: born at 32 weeks, 1,300 g. Baby Z: born at 36 weeks, 3,200 g. Then write which baby is near the 50th line, which is below the 10th, and which is above the 90th. (4 pts)

Answer: Three dots in roughly the right places (2 pts; 1 pt off per misplaced dot). X ~ 50th (typical); Y below the 10th (small for gestational age); Z above the 90th (large for gestational age) (2 pts).

Part D — Following the curve

5. Baby M was born at 30 weeks weighing 1,450 g — right on the 50th line. Two weeks later, at 32 weeks, she weighs 1,450 g. (The 50th line at 32 weeks is about 1,800 g; the 10th is about 1,400 g.) What has happened to her place in line? Is this an alarm, even though 1,450 g is still 'a normal weight'? What would the team probably do? (3 pts)

Answer: She hasn't grown for two weeks, so she's dropped from the 50th to about the 10th — crossing curves downward, 'falling off the curve' (1 pt). Yes, it's an alarm: the dot looks fine but the line is flat (Calculus Week 1) (1 pt). The team looks for a reason (breathing hard, infection, not absorbing feeds) and adds calories — more milk, or fortifier (1 pt).

Part E — Flashbacks

6. From NICU Tutor: a baby born weighing 1,300 g is called what kind of low birth weight? (1 pt)

Answer: Very low birth weight (under 1,500 g). Under 2,500 is low; under 1,000 is extremely low.

7. From last week's Statistics: what is the mean of 110, 120, 130, and 160? (1 pt)

Answer: $520 \div 4 = 130$.

Part F — Think like a doctor

8. Two babies in the follow-up clinic are both at the 5th percentile for weight today. Baby P has been at the 5th percentile at every visit since birth; both parents are small. Baby Q was at the 50th percentile a month ago. Which baby worries you, and why? What does this tell you about what a percentile can and can't say? (3 pts)

Answer: Baby Q (1 pt): she has crossed from the 50th down to the 5th — falling off her curve — while P is following hers. A percentile is a place in line, not a grade; a low number that's stable can be perfectly healthy (small parents, small baby), while a CHANGE in percentile is the real signal (1 pt). For Q the team hunts for a cause and adds calories; for P, keep watching and reassure the parents (1 pt).

HOME LAB (optional, not graded): Your percentile in the family

You need: a ruler or tape measure and everyone at home (plus friends or neighbors if you can catch them — more people makes a better line)

1. Measure every person's hand span: thumb tip to pinky tip, fingers spread wide. Write each one down with the person's name.
2. Line the numbers up from smallest to biggest. Find the median (the middle one, or halfway between the two middles).
3. Find YOUR position in line. Percentile = your position $\div$ number of people $\times$ 100.
4. Now measure heights and do it again. Did your percentile change?
5. Bonus: add a grown-up's shoe or a stuffed animal's paw and watch what happens to the mean but not the median.

What you should see: You'll get a median hand span and a percentile for yourself — probably a low one, since grown-ups have bigger hands. Your height percentile is usually different: a percentile depends on who is in the line. Adding one wild number moves the mean but not the median.

